

An LUR Non-UR Ball Without the Uniformly Continuous Quasiconvex Extension Property

Candidate partial result for arXiv:2603.05206

fa_banach_001, lane 14, agent_lane_14

17 August 2026

Status. Candidate partial result, likely valid, pending human review. The theorem below gives an explicit LUR but non-UR body for which the extension property fails. It does *not* prove the source’s universal necessity statement for every LUR non-UR body.

1 Source question and scope

The source is C. A. De Bernardi and L. Veselý, *Extending quasiconvex functions from uniformly convex sets*, arXiv:2603.05206 (2026) [1]. On page 2, after recording that uniform rotundity (UR) is sufficient and failure of local uniform rotundity (LUR) is an obstruction, the authors ask:

Is the UR property of a convex body C necessary for every uniformly continuous quasiconvex function on C to extend to a uniformly continuous quasiconvex function on the whole ambient space? Equivalently, what happens for bodies which are LUR but not UR?

It is natural to ask whether some analogous result holds for QC functions. It turns out that the “quasiconvex” situation is completely different and depends heavily on geometric properties of the convex set C . Here we mainly consider the case in which C is a convex body. In the following theorem, we collect our main previous results, contained in [5, 6]. The natural definitions of a UR (uniformly rotund) body and an LUR (locally uniformly rotund) body can be found in the cited papers.

Theorem 1.2. *Let X be a real normed space of dimension at least two.*

- (a) *There exists a Lipschitz QC function f on the closed unit disc $D \subset \mathbb{R}^2$ which does not admit any Lipschitz QC extension to the whole $\mathbb{R}^2$ (see [5, Example 4.2]). Consequently,¹ X contains a convex body C such that some Lipschitz QC function on C cannot be extended to a Lipschitz QC function on the whole X .*
- (b) *If X is a Banach space and $C \subset X$ is a convex body which is not LUR, there exists a Lipschitz QC function $f: C \rightarrow \mathbb{R}$ which does not admit any continuous QC extension to the whole X (see [5, Theorem 3.3]).*
- (c) *If $C \subset X$ is a UR convex body, then each uniformly continuous QC function $f: C \rightarrow \mathbb{R}$ admits a uniformly continuous QC extension to X (see [6, Theorem 5.5]).*

It remains an *open problem* whether the UR property of a convex body C is also necessary for extendability of each uniformly continuous QC function on C to a uniformly continuous QC function on the whole X . In other words, we don't know what happens for convex bodies which are LUR but not UR.

However, in the finite-dimensional case, we obtain exact characterizations of various types of extendability in terms of geometric properties of C . And these results are the subject of the present paper.

The paper is organized as follows. After a brief section of preliminaries, the next two Sections 3 and 4 provide three (finite-dimensional) constructions of Lipschitz QC functions that cannot be extended to sufficiently good QC functions defined on the whole space (Theorems 3.1, 3.2, and 4.1). In Section 5, we show that on every convex body in an arbitrary normed space of dimension at least 2 there exists a Lipschitz QC function which has no Lipschitz QC extension to the whole space (Theorem 5.2). Section 6 contains some technical

We show that the desired extension property already fails for a concrete LUR non-UR unit ball. Thus LUR alone is not sufficient. What remains open is whether *every* non-UR body admits such a witness.

2 Definitions

A function f on a convex set is *quasiconvex* (QC) if each sublevel $\{f \leq r\}$ is convex. A norm, or its closed unit ball, is *locally uniformly rotund* if, whenever x, y_j are unit vectors and $\|x + y_j\| \rightarrow 2$, one has $\|x - y_j\| \rightarrow 0$. It is *uniformly rotund* if the latter implication is uniform in the base point: for every $\delta > 0$ there is $\eta > 0$ such that unit vectors separated by at least δ have midpoint norm at most $1 - \eta$.

3 Idea of the proof

The proof of Theorem 3.2 in the source paper constructs a Lipschitz QC function whose increasing strict sublevels are a fan of half-planes converging to a flat boundary segment. We put this same fan into a sequence of increasingly square but strictly convex two-dimensional balls $B_{\ell_{p_n}^2}$, where $p_n \rightarrow \infty$. Their ℓ_2 -sum is LUR because every individual block is LUR, but it is not UR because the blocks become arbitrarily flat.

On the sum ball we take the supremum of the coordinate copies of the planar witness. The coordinate structure does two jobs: it keeps this supremum quasiconvex, and it lets us recover the full planar witness by restriction to one block. Convexity then forces any extension's strict sublevel in that block to remain on one side of the corresponding cutting line. Across all blocks this forces an order-one jump between pairs whose distances tend to zero, contradicting uniform continuity.

4 The planar input

Let $D = [-1, 1]^2$, $z = (0, 1)$, and, for $k \geq 1$, put

$$\varepsilon_k = 2^{-k}, \quad b_k = 1 - 2^{-k}, \quad L_k(t) = 1 + \varepsilon_k - \frac{\varepsilon_k}{b_k}t,$$

and $H_k = \{(t, s) : s < L_k(t)\}$. The following is the precise piece of the source construction that we use.

Lemma 1 (normalized planar fan). *There are a Lipschitz QC function $q : D \rightarrow [0, 1]$ and numbers $0 < \tau_2 < \tau_3 < \dots < 1$, with $\tau_k \rightarrow 1$, such that*

$$q(0, 0) = q(z) = 0 \quad \text{and} \quad \{q < \tau_k\} \cap B_{\ell_p^2} = H_k \cap B_{\ell_p^2} \quad (4.1)$$

whenever $1 < p < \infty$ and the line $s = L_k(t)$ crosses $\text{int } B_{\ell_p^2}$.

Proof. This is a normalized restriction of the construction in the proof of [1, Theorem 3.2]. After a linear rescaling of its flat boundary segment, that proof uses the open half-planes H_k inside $A = \{s < 1\}$ and supplies a Lipschitz QC g on $\bar{A}$ and increasing levels α_k satisfying

$$\{g < \alpha_k\} \cap A = H_k \cap A, \quad g(0, 0) < \alpha_1, \quad g(z) \leq \alpha_1.$$

The construction comes from [2, Lemma 2.9]; the source proof also notes $\alpha_1 < \alpha_\infty := \sup_k \alpha_k$.

Here $\alpha_\infty < \infty$. Indeed, for every k the point $t = 1$ admits an $s \in (L_k(1), 1)$, hence a point of the compact square $D \cap A$ outside H_k . At that point $g \geq \alpha_k$, so $\alpha_k \leq \max_D g$. The levels are strictly increasing because their strict sublevels $H_k \cap A$ are distinct. Define

$$q = \min \left\{ 1, \max \left\{ 0, \frac{g - \alpha_1}{\alpha_\infty - \alpha_1} \right\} \right\} \Big|_D, \quad \tau_k = \frac{\alpha_k - \alpha_1}{\alpha_\infty - \alpha_1}.$$

A nondecreasing scalar composition preserves quasiconvexity, and the clamp preserves Lipschitz continuity. Thus q has the asserted regularity and low values. Away from the top edge $s = 1$, its strict τ_k -sublevel is exactly H_k . A strictly convex ℓ_p^2 ball meets that top edge only at z ; since $q(z) = 0 < \tau_k$ and $z \in H_k$, equality (4.1) follows. Finally $\tau_k \rightarrow 1$ by definition of the supremum. $\square$

5 Two elementary geometric lemmas

Lemma 2 (the cutting lines enter explicit blocks). *For $n \geq 1$, set $k = n + 1$ and $p_n = 2^{3k}$. The line $s = L_k(t)$ crosses $\text{int } B_{\ell_{p_n}^2}$.*

Proof. Write $\varepsilon = 2^{-k} \leq 1/4$ and take

$$t = \frac{b_k + 1}{2} = 1 - \frac{\varepsilon}{2}, \quad s = L_k(t) = 1 - \frac{\varepsilon^2}{2(1 - \varepsilon)}.$$

Both coordinates lie strictly between zero and one. Since $1 - u \leq e^{-u}$, and $p_n = \varepsilon^{-3}$,

$$t^{p_n} + s^{p_n} \leq \exp\left(-\frac{1}{2\varepsilon^2}\right) + \exp\left(-\frac{1}{2\varepsilon(1-\varepsilon)}\right) \leq e^{-8} + e^{-8/3} < 1.$$

Thus (t, s) is an interior point of the block ball lying on the line. $\square$

Lemma 3 (half-plane continuation). *Let K be a convex body in $\mathbb{R}^2$, let H be an open half-plane whose boundary line meets $\text{int } K$, and let U be open and convex. If $U \cap K = H \cap K$, then $U \subset H$.*

Proof. Write $H = \{\ell < 0\}$ for an affine functional ℓ , and choose $x \in \{\ell = 0\} \cap \text{int } K$. If U contained a point y with $\ell(y) \geq 0$, openness would give a nearby $y' \in U$ with $\ell(y') > 0$. Choose $u \in H \cap \text{int } K$ arbitrarily close to x . Then $u \in U$. The segment $[u, y']$ lies in U , and, by taking u sufficiently close to x , a point of this segment just beyond the line remains in $\text{int } K$. It belongs to $U \cap K$ but not to $H \cap K$, a contradiction. $\square$

6 Main theorem

Theorem 1. *There exist a separable Banach space X , an LUR but non-UR closed unit ball $C = B_X$, and a Lipschitz quasiconvex function $f : C \rightarrow \mathbb{R}$ which has no uniformly continuous quasiconvex extension to X .*

Proof. For $p_n = 2^{3(n+1)}$ let $E_n = \ell_{p_n}^2$ and put

$$X = \left(\bigoplus_{n=1}^{\infty} E_n \right)_2, \quad \|(x_n)\|_X = \left(\sum_n \|x_n\|_{p_n}^2 \right)^{1/2}.$$

This space is separable.

The ball is LUR. Every finite-dimensional ℓ_p^2 , $1 < p < \infty$, is LUR. For completeness, the ℓ_2 -sum preserves this property as follows. Suppose $x = (x_n)$ and $y^j = (y_n^j)$ are unit vectors and $\|x + y^j\| \rightarrow 2$. The vectors of component norms

$$a = (\|x_n\|) \quad \text{and} \quad c^j = (\|y_n^j\|)$$

are unit vectors of ℓ_2 , and $\|x + y^j\| \leq \|a + c^j\|_2 \leq 2$. Uniform convexity of ℓ_2 gives $c^j \rightarrow a$. Moreover,

$$\sum_n \left[(a_n + c_n^j)^2 - \|x_n + y_n^j\|^2 \right] \rightarrow 0.$$

For each fixed n with $a_n > 0$, the n th nonnegative summand tends to zero; after normalizing in E_n , LUR gives $y_n^j \rightarrow x_n$. If $a_n = 0$, the same conclusion follows from $c_n^j \rightarrow 0$. Coordinate convergence, together with $c^j \rightarrow a$ and a finite-head/small-tail argument, yields $\|y^j - x\|_X \rightarrow 0$.

The ball is not UR. In the n th block consider the unit vectors

$$u_n = (0, 1), \quad v_n = \left(\frac{1}{2}, (1 - 2^{-p_n})^{1/p_n} \right).$$

Their distance is at least $1/2$. The second coordinate of their midpoint tends to one, so by convexity

$$1 \geq \left\| \frac{u_n + v_n}{2} \right\|_{p_n} \geq \frac{1 + (1 - 2^{-p_n})^{1/p_n}}{2} \rightarrow 1.$$

Embedding these vectors into the n th coordinate of X violates uniform rotundity.

The function. Let q be supplied by Lemma 1 and define, for $x = (x_n) \in C$,

$$f(x) = \sup_n q(x_n).$$

The square contains every block ball. If q has Euclidean Lipschitz constant M , then

$$|q(r) - q(s)| \leq M\|r - s\|_2 \leq \sqrt{2}M\|r - s\|_{p_n}$$

for all n . Consequently

$$|f(x) - f(y)| \leq \sqrt{2}M \sup_n \|x_n - y_n\|_{p_n} \leq \sqrt{2}M\|x - y\|_X.$$

Also $\{f \leq r\} = C \cap \prod_n \{q \leq r\}$ is convex, so f is QC. Because $q(0) = 0$ and $q \geq 0$, the restriction of f to the n th coordinate ball is exactly q .

No uniformly continuous QC extension. Assume that $F : X \rightarrow \mathbb{R}$ is such an extension. In the n th coordinate plane set

$$U_n = \{w \in E_n : F(w) < \tau_{n+1}\}.$$

This is open and convex, and Lemma 1 gives

$$U_n \cap B_{E_n} = H_{n+1} \cap B_{E_n}.$$

By Lemma 2, the cutting line meets the interior of the block ball. Lemma 3 therefore yields $U_n \subset H_{n+1}$.

Now regard the following as points in the n th coordinate plane:

$$z_n = (0, 1), \quad a_n = (0, 1 + \varepsilon_n).$$

At $t = 0$, the boundary of H_{n+1} has height $1 + \varepsilon_{n+1} < 1 + \varepsilon_n$, so $a_n \notin H_{n+1}$ and hence

$$F(a_n) \geq \tau_{n+1} \longrightarrow 1.$$

On the other hand, $z_n \in C$ and $F(z_n) = f(z_n) = q(z) = 0$, while

$$\|a_n - z_n\|_X = \varepsilon_n \longrightarrow 0.$$

This contradicts uniform continuity of F . □

7 Verification and limitations

The script `code/verify_geometry.py` checks the first twenty explicit blocks in log space: the point in Lemma 2 is strictly inside the block ball, the analytic exponential upper bound is below one, a_n is beyond the next cutting line, and the non-UR chord has fixed separation. These computations are not used as proof.

Five focused upgrades were attempted: moving fans along arbitrary bad chords; supremum amalgamation without coordinates; projection and quotient reductions; an ultrapower reduction; and Baire/diagonal uniformization of extension moduli. The common obstruction is localization. In an arbitrary LUR non-UR body the bad chords move, and no operation was found that combines their local witnesses while preserving both quasiconvexity and uniformly low values at all comparison

points. The ℓ_2 sum solves exactly this problem by coordinate isolation. Thus Theorem 1 is a proper partial result and the universal source question remains open.

A bounded local-registry and arXiv/web search on 17 August 2026 used the exact source title and the phrases “quasiconvex extension LUR uniformly rotund,” “quasiconvex extension locally uniformly rotund,” and “uniformly continuous quasiconvex extension.” It recovered the source, the authors’ earlier positive UR theorem [3], and general LUR/non-UR renorming literature, but no later answer or matching block-sum construction. This is not an exhaustive priority search; novelty confidence is moderate.

Human-review recommendation. Check the normalized planar-fan extraction in Lemma 1, especially the behavior at the top-edge point z , and the use of open strict sublevels in Lemma 3. The remaining block-sum argument is elementary and self-contained.

References

- [1] C. A. De Bernardi and L. Veselý, *Extending quasiconvex functions from uniformly convex sets*, arXiv:2603.05206 (2026).
- [2] C. A. De Bernardi and L. Veselý, *Rotundity properties, and non-extendability of Lipschitz quasiconvex functions*, J. Convex Anal. **30** (2023), 329–342.
- [3] C. A. De Bernardi and L. Veselý, *On extension of uniformly continuous quasiconvex functions*, Proc. Amer. Math. Soc. **151** (2023), 1705–1716.